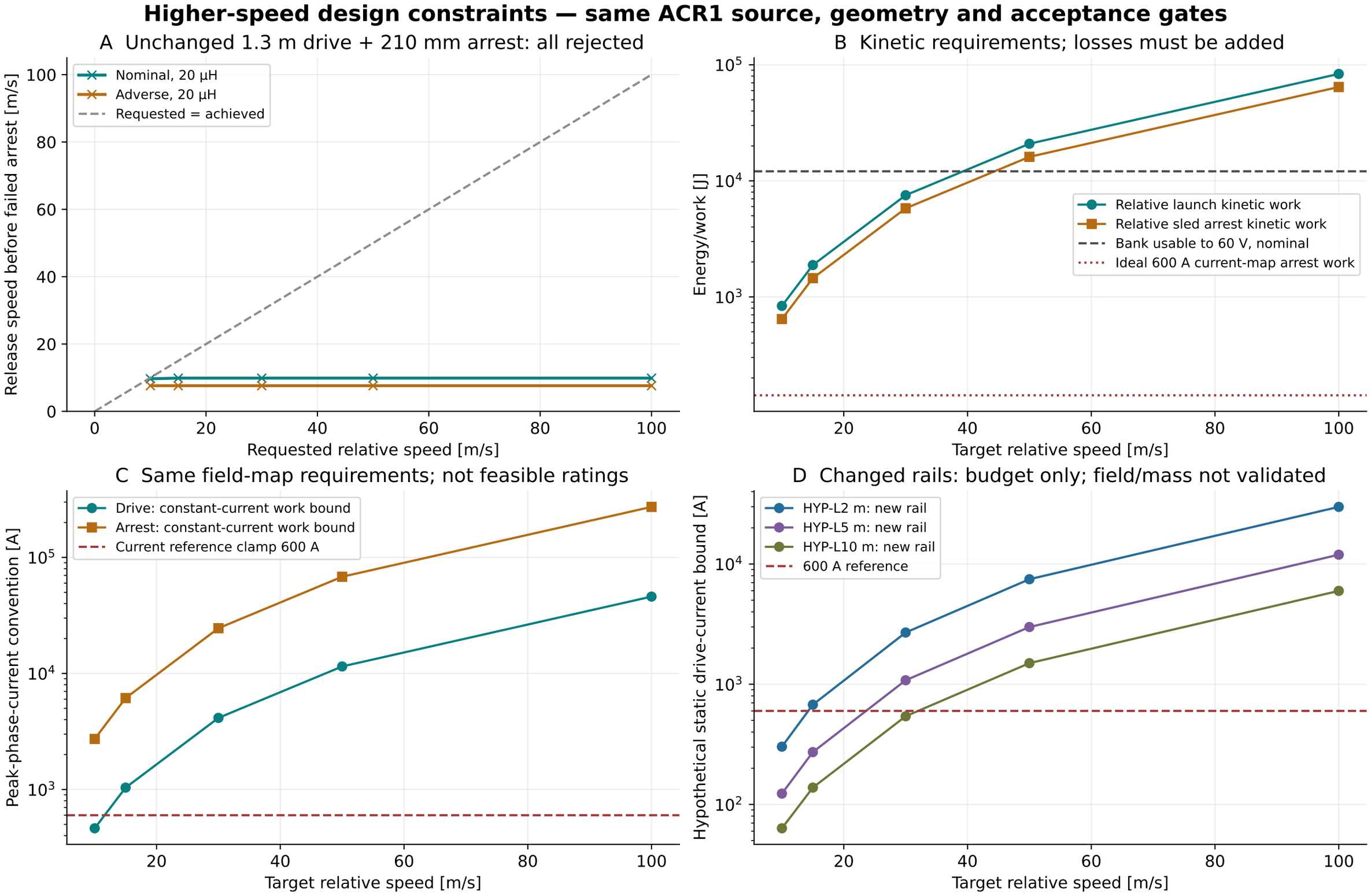
Higher-speed design constraints — same ACR1 source, geometry and acceptance gates
A Unchanged 1.3 m drive + 210 mm arrest: all rejected
Nominal, 20 µH
Adverse, 20 µH
Requested = achieved
Release speed before failed arrest [m/s]
Requested relative speed [m/s]
B Kinetic requirements; losses must be added
Relative launch kinetic work
Relative sled arrest kinetic work
Bank usable to 60 V, nominal
Ideal 600 A current-map arrest work
Energy/work [J]
Target relative speed [m/s]
C Same field-map requirements; not feasible ratings
Drive: constant-current work bound
Arrest: constant-current work bound
Current reference clamp 600 A
Peak-phase-current convention [A]
Target relative speed [m/s]
D Changed rails: budget only; field/mass not validated
HYP-L2 m: new rail
HYP-L5 m: new rail
HYP-L10 m: new rail
600 A reference
Hypothetical static drive-current bound [A]
Target relative speed [m/s]